

detect an effect of ozone on the risk of death from cardiovascular causes when the concentration of PM_{2.5} was taken into account. But they did demonstrate a significant increase in the risk of death from respiratory causes in association with an increase in ozone concentration.

Mortality, Respiratory (ICD-9 code 460-519)

In single-pollutant models, the coefficient and standard error are estimated from a relative risk of 1.04 (95% CI: 1.03-1.07) for females and 1.01 (95% CI: 0.99-1.04) for males for a 10 ppb change in ambient ozone concentration measured from April to September during the years from 1977 to 2000 (Jerrett et al. 2009, Table 4).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.7.3 Katsouyanni et al. (2009)

See full study description under Katsouyanni et al. (2009) in Appendix F, Section F.1.1.

Short-term Mortality, All Cause

In single pollutant models, the coefficient and standard error are based on the summer-only penalized spline estimate of all-cause mortality of 0.70% (0.28, 1.12%) for ages <75 and 0.62% (0.16, 1.08%) for ages ≥75 per 10 µg/m³ (equivalent to 5.1 ppb) increase in O₃ from distributed lag days (Katsouyanni et al. 2009, Table 24: Distributed Lags; Penalized splines; O₃ Results).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.7.4 Lin et al. (2005)

Lin et al. (2005) examined the relationship between short term O₃ exposures and hospital admissions for lower respiratory infections (ICD-9 464, 466, 480-487) in a case-crossover study among Toronto residents under the age of 15 between 1998 and 2001. Data on 6,782 hospitalizations were obtained from the Discharge Abstract Database. Daily 1-hour maximum O₃ concentrations in all seasons were obtained from seven monitoring stations in the National Air Pollution Surveillance system. The authors ran conditional logistic regression models controlling for temperature and weather conditions using 1- to 7-day average lags.

Hospital Admissions, All Respiratory

In multi-pollutant models adjusted for PM_{2.5} and PM_{10-2.5}, the coefficient and standard error are estimated from an odds ratio of 1.18 (95% CI: 0.94-1.47) for females and 1.08

(95% CI: 0.89-1.31) for males for a 21.8 ppb increase in daily 1-hour maximum O₃ concentrations with a 4-day average lag (Lin et al. 2005, Table 3 (Adjusted B)).

The Health Impact Function was adjusted from the daily 1-hour max metric to the daily 8-hour max metric using a ratio of 1.13 (ratio of 1-hour max to 8-hour max ozone) (Anderson and Bell 2010, Table 2).

F.7.5 Mar and Koenig (2009)

Mar and Koenig (2009) evaluated the relationship between outdoor ozone in the summer and asthma aggravation. The authors used hospital data on daily asthma cases from 1998 to 2002 in Seattle with local monitored PM_{2.5} and ozone concentrations to assess the association between asthma visits to the emergency department and air pollutants. They analyzed 1-hour and 8-hour max ozone concentrations at two monitors in Greater Seattle. Asthma ED visits were analyzed at 0 through 5-day lags. The authors found that ozone exposure exacerbates asthma in people in the Seattle area, especially in children. Authors found that in adults during the warmer months between May and September, a 10 ppb increase in 8-hour maximum ozone concentration is associated with relative risk of asthma-related ED visits of 1.08 (1.02, 1.14) with a 4-day lag. In children, during the warmer months, a 10 ppb increase in 8-hour maximum ozone concentration is associated with relative risk of asthma-related ED visits of 1.11 (1.02, 1.21) with a 3-day lag. The difference in lag and relative risk between children and adults suggests that children are more immediately responsive to the adverse effects of ozone exposure.

Emergency Room Visits, Asthma

In single-pollutant models, the coefficient and standard error are estimated from a relative risk of 1.11 (95% CI: 1.02-1.21) for age <18 and 1.04 (95% CI: 0.99-1.10) for age ≥18 for a 10 ppb increase in daily 8-hour maximum summer ozone concentration with a 3-day average lag (Mar and Koenig, 2009, Table 5 and Table 6).

F.7.6 Medina-Ramon & Schwartz (2008)

Medina-Ramon & Schwartz (2008) evaluated short-term O₃ exposure and all-cause mortality among residents of all ages in 48 U.S. cities from 1989-2000 using a case-only approach. Data on 2,729,640 non-accidental deaths was obtained from the National Center for Health Statistics. The authors estimated 8-hour daily mean ozone concentrations (warm season, May to September) for each city using daily ozone levels reported by the U.S. EPA Aerometric Retrieval System. The authors ran city-specific conditional logistic regressions controlling for seasonality, temperature, and day of the week and pooled the results across cities. Results were presented by population subgroup characteristics and chronic conditions.

Mortality, All Cause

The O₃-mortality risk estimates for at-risk subpopulations reported in Medina-Ramon & Schwartz (2008) required additional modification in order to use those results to

develop health impact functions. The authors presented excess risk estimates for each subpopulation as the additional percent change in mortality for persons who have the condition, compared to persons without the condition. For our populations of interest, these subgroups were persons age 65 or older compared to those younger than 65, and females relative to males. However, they did not report the risk estimate for these comparison groups, so in order to estimate the total excess risk for each exposed at-risk group, we needed to first back-calculate the excess risk for the comparison group without the factor of interest. We accomplished this by assuming that the authors' overall reported excess risk for the full sample of 0.65% (95% confidence interval = 0.38% to 0.93%) could be expressed as a weighted average of the unreported excess risk ("x") and the full excess risk for the at-risk group, which would be expressed as the sum of x and the reported excess risk from Medina-Ramon & Schwartz (2008) Table 2, where the weights are calculated using the total and at-risk group sample sizes in Table 1 of that paper. For example, to calculate the total excess risks for the females in the sample, we used the following equation:

$$ER_{Total} = \frac{ER_{Male}(Pop_{Male}) + ER_{Female}(Pop_{Female})}{Pop_{Total}}$$

where ER_{Total} is the full sample excess risk of 0.65%; ER_{Female} is the excess risk of ozone exposures for females; ER_{Male} is the excess risk of ozone exposures for males; Pop_{Total} is the total sample population; and Pop_{Female} and Pop_{Male} are the size of the female and male subsets of the sample population, respectively. We also know from Table 2 of that paper that:

$$ER_{Female} = ER_{Male} + 0.58 \%$$

Substituting and using the available information from Medina-Ramon & Schwartz (2008) Tables 1 and 2, we can solve for ER_{Male} and then ER_{Female} :

$$0.65\% = \frac{ER_{Male}(1,365,937) + (0.58\% + ER_{Male})(1,363,703)}{2,729,640}$$

$$ER_{Male} = 0.36 \%$$

and

$$ER_{Female} = 0.36\% + 0.58\% = 0.94\%$$

We then used the full excess risk value for the female subpopulation to derive a health impact function for ozone-related mortality for females. Final calculated excess risks are 0.94% (0.47-1.42%) for females aged 0-99; 0.36% (0.29-0.44%) for males aged 0-99, -0.13% (-0.33-0.07%) for both sexes aged 0-64, and 0.97% (0.51-1.44%) for both sexes aged 65-99.

F.7.7 Paulu and Smith (2008)

Paulu and Smith (2008) conducted a case-crossover analysis to evaluate the relationship between daily ozone concentrations and emergency room visits for asthma (ICD-9 493) among Maine residents aged 2 and older from 2000 to 2003. Data on 8,020 asthma-related ER visits was obtained from the Maine Health Data Organization. Daily 8-hour maximum O₃ concentrations were computed from two monitor sites situated in or near Portland, ME and data was obtained from the Maine Department of Environmental Protection, Bureau of Air Quality. The authors defined the warm season as May 22-September 10 (2000) and May 23-September 11 (2001-2003). The authors ran conditional logistic regression models stratified by sex and age groups (2-14, 15-34, 35-64, and ≥65 years) controlling for temperature, humidity, and day after major holidays as well as PM_{2.5} in co-pollutant models. Paulu and Smith (2008) found that excess risk was concentrated among females aged 15 to 34 and males younger than 15.

Emergency Room Visits, Asthma

In single-pollutant models for both sexes, the coefficient and standard error are estimated from an excess risk of 11% (1-23%) for ages 2-14 and 16% (8-24%) for ages 15-34 for a 10 ppb increase in average daily 8-hour maximum ozone (lags 0-3 days) (Paulu and Smith, 2008, Figure 1 text).

In single-pollutant models for ages 2 and above, the coefficient and standard error are estimated from an excess risk of 12% (6-18%) for females and 11% (4-18%) for males for a 10 ppb increase in average daily 8-hour maximum ozone (lags 0-3 days) (Paulu and Smith, 2008, Figure 1 text).

F.7.8 Villeneuve et al. (2007)

Villeneuve et al. (2007) evaluated the relationship between short-term ozone exposure and emergency room visits for asthma (ICD-9 493) among residents of Edmonton, Canada aged 2 and above from 1992 to 2002 in a case-crossover study. Data on 57,912 asthma-related ER visits was provided by Capital Health. Daily 8-hour maximum O₃ concentrations were obtained from automated fixed-site monitoring stations maintained by Environment Canada as part of the National Air Pollution Surveillance Network. The monitors measured both warm season (April-September) and cold season (October-March) ozone concentrations. The authors ran conditional logistic regression models controlling for temperature, humidity, influenza, and aeroallergens. Villeneuve et al. (2007) found associations between ozone and asthma emergency room visits in the warm season and observed the strongest effects in young children.

Emergency Room Visits, Asthma

In single-pollutant models, the coefficient and standard error are estimated from odds ratios of 1.06 (0.94-1.19) for ages 2-4; 1.14 (1.05-1.24) for ages 5-14; 1.11 (1.04-1.18) for ages 15-44; 1.12 (1.00-1.26) for ages 45-64; 1.14 (0.94-1.39) for ages 65-74; and 0.99 (0.78-1.25) for ages 75-99 for a 18 ppb increase in average daily 8-hour maximum warm season ozone (5-day average lag) (Villeneuve et al., 2007, Tables 4, 5, 6, 7, 8, 9).